

Osaid Khan

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EDUCATION

Pace University

Master of Science (MS) in Computer Science (GPA - 4.0/4.0)

New York City, NY

Jan 2024 - Dec 2025

Dr. A.P.J. Abdul Kalam Technical University

Bachelor of Technology (B.Tech) in Computer Science

Noida, India

Jul 2016 - Sep 2020

TECHNICAL SKILLS

Languages: Java, JavaScript/TypeScript, Rust, Python

Frameworks: FastAPI, FastMCP, LangGraph, LangChain, LangSmith, PyTorch, React, Sprint Boot

Databases: PostgreSQL, MongoDB, Redis

AWS Services: EC2, RDS, S3, Lambda, CloudWatch, Step Functions, CloudFormation, CloudFront, API Gateway

AI: Claude code, Codex, Pi coding-agent

Others: Google Cloud, Git, CI/CD, Kafka, Node.js

PROFESSIONAL EXPERIENCE

Software Engineer

Phoenix, AZ

Spense

Feb 2026 - present

- Designed a **hierarchical multi-agent system** for a multi-tenant **CRM SaaS** application using **LangGraph**, **LangChain** and **FastAPI** with a main orchestrator lazily loads **sub-agent catalogs** and delegates to over **40 specialized sub-agents**, each built per request with the user's **JWT**, **tenant and role context** propagated end-to-end so every backend call respects the caller's permissions
- Compressed **500 backend operations** into **tool-group dispatchers** allowing progressive disclosure of tools collapsing 60+ REST endpoints per agent into a few typed entry points, **lifting correct tool selection 90%** and cutting **sub-agent context bloat** and tool-call errors
- Built **trace-driven eval suites** that replay real production conversations against new prompts and model versions, with **real-time monitors** catching agents going off the rails before they reach customers
- Added **retrieval-augmented agents** for in-product how-to questions and a **coding agent** that accelerates customer onboarding by **60%** by generating ready-to-run code examples directly from the CRM API docs so customers can build custom UIs and integration flows without leaving the product, protected by **two-stage input safety** (jailbreak regex + **Lakera Guard**) with **SQLite checkpointing** for multi-turn state; separately built a **FastMCP** server exposing **1000+ typed tools** for external AI clients, every call traced in **Langfuse**

AI Researcher

New York City, NY

Pace Artificial Intelligence Lab

Sep 2025 - Dec 2025

- Researched **agent evaluation** on Pace's **HPC cluster**, combining multiple weak judges (**LLM-as-judge** + reward models) into a single strong evaluator that ranked agent runs on multi-step tool-use tasks with near-expert accuracy
- Built an **eval harness** that scored **LangGraph** agent runs with a evaluator and fed the top runs back into prompt tuning, lifting **tool-call success** by **60%** over a single **LLM-judge** baseline

Software Engineer

Noida, India

Qualitest

Feb 2021 - Dec 2023

- Built **RAG AI chatbot** for a **K-12 education platform** using **LangChain** and **PostgreSQL pgvector** store, **improving customer engagement by 50%** by enabling natural-language queries over course content and platform documentation
- Designed and built REST APIs using **Golang** and **Spring Boot**, containerized on **GCP Kubernetes (GKE)**, with **OpenTelemetry** instrumentation and custom Grafana dashboards tracking **p95/p99 API latencies**

PROJECTS

Dark Factory - Distributed orchestrator for PI coding agent | *Elixir, TypeScript* github.com/ozzyozbourne/dark-factory

- Turns a single user goal into a **dependency-aware task plan** that is automatically dispatched to a **fleet of AI coding agents running across multiple machines**, with built-in retries and timeout handling
- Built a **fault-tolerant Elixir coordinator** that matches tasks to agents by capability and live workload over a **real-time WebSocket protocol**, with a **CRDT-backed distributed registry** keeping the cluster alive when individual nodes fail